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An atom-photon quantum gate using an atomic clock qubit and a birefringent cavity

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Abstract

Single atoms coupled to optical cavities provide a powerful platform for photonic quantum information processing. Minimizing the influence of external factors like magnetic field fluctuations is a key goal in this endeavor. In this work, I investigate a novel gate protocol that employs a single ^{87}Rb atom prepared in a magnetic-field-insensitive clock qubit and coupled to a high-finesse birefringent FabryPerot cavity. The cavitys high birefringence creates two well-separated polarization eigenmodes, enabling polarization-selective reflectivity that depends on the atomic qubit state. This allows for the implementation of a controlled phase (CPHASE) gate between the atomic clock states and our photonic qubits. By rotating the basis of the photonic qubit, one is then able to realize an atom-photon controlled-NOT (CNOT) gate.

Show some results here and talk about how well it works and also how one might be able to use the gate

I present simulations based on inputoutput theory and master-equation modeling that quantify the conditional reflection amplitudes, gate truth table, and resulting fidelities under realistic experimental imperfections such as finite mode matching, cavity loss channels, and multiphoton contributions. Our results indicate that atom-state-dependent phase shifts on the photonic qubit are achievable in the current system, providing a viable path toward a robust, high-fidelity, cavity-assisted atomphoton quantum gate.

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Chapter 1

Introduction

Few theories have reshaped modern technologies and thus our day to day life as profoundly as quantum mechanics when it was discovered in the early 1900s. The examination of the black-body radiation of celestial bodies such as our sun and later formulation of the Planck's law by Max Planck, explaining the quantized energy exchange between a black-body and a radiation field, laid the ground work for many theories and technologies. Among these is the laser which is based on Albert Einstein's theory on stimulated emission [1]. The laser has found its usage both in day-to-day life (e.g. the barcode scanner at the register, or scanner within a computer mouse) as well as research (e.g. Sisyphus cooling [2] or trapping of atoms in a magneto optical trap [3]). Another technology of great importance is the atomic clock, which is based on Norman Ramsey's work [4], allowing for the clear definition of a single second [5]. One piece of technology, which shapes our modern civilization the most, is that of the transistor, specifically the metaloxidesemiconductor field-effect transistor (MOSFET). The MOSFET transistor is the most used type of transistor found in modern electronics [6] that either allows or denies the passage of electrical current. By either blocking the current or letting it pass through the component, one is able to essentially construct a miniaturized switch, that can be used to realize a single *bit*. Interestingly enough, the theory which explains the passage of current through such a device, that being the electronic band structure model, is again rooted in quantum mechanics.[7]

In an attempt to predict the following years of chip evolution, Gordon Moore made the bold assumption in 1965 that with each coming year, the number of components crammed on an integrated chip will double [8]. This would later on be known as *Moore's Law*, and would pave the way for chip manufacturers to keep on increasing the amount of transistors in their chips, in order to subconsciously keep up with the predictions made by Moore. Back in the day, with the way technology was evolving, this enthusiasm towards simply increasing the amount of transistors by following a "plenty of room at the bottom"-mentality from the likes of Richard Feynman [9] and thus achieving an increase in computing performance, would eventually hit fundamental limits. In order to pack more transistors onto the same sized computer chip, the transistors themselves would need to be shrunk down. However in recent years, this approach of increasing computing performance has led to the realization that simply going smaller, does not yield the hoped-for performance increase. [10]

As the tasks one concerns themselves with become much more intricate, like finding the most efficient delivery route for a logistics company[11], or accurately simulating small subsystems [12, 13, 14], it is clear that classical computation has hit a limit and a new kind of computation platform is needed. As Feynman points out in his talk "Simulating Physics with Computers":

> “Nature isn't classical, dammit, and if you want to make a simulation of nature, you'd better make it quantum mechanical, and by golly it's a wonderful problem, because it doesn't look so easy.”[12]

So while classical computers were still trying to realize the trend Moore predicted in 1965, the foundation for a new paradigm of computation were already being laid. In 1980, Paul Benioff proposed the first quantum mechanical model for a Turing Machine[15], demonstrating that computation could in principle be carried out using quantum mechanical processes. This new way of computing is called *quantum computing*, which instead of using fixed bits that can either be a 1 or a 0, uses the so called *qubits*. These qubits are in itself a two-state quantum-mechanical system, and can be placed in any *superposition* state $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$. That is, until a measurement is performed and the state collapses back to being either $|0\rangle$ or $|1\rangle$ again. What makes the qubit interesting however, is the relationship multiple qubits can have with one another, and that they can be *entangled* with one another, thus not allowing for description of one qubit state, without the state of another qubit. By leveraging these two fundamental quantum mechanical principles, many protocols have been implemented that in theory should outperform their classical counterparts. Be it in the realm of encryption where certain quantum algorithms now promise an exponential speedup [16], in quantum key distribution where the laws of quantum mechanics enable fundamentally secure communication [17], or in the transfer of quantum states between distant nodes via quantum teleportation [18].

However, as time went on, and different hardware platforms were investigated that could serve as a foundation for a quantum computer, one problem (along many others), became apparent: scalability. As one tries to increase the amount of qubits used in a quantum computer, being able to keep external noise and errors that each component produces at bay, becomes more and more of a challenge[19]. In order to be able to scale these new systems up, it was proposed that instead of building one large quantum computer, where one would try to keep errors at bay via error correction algorithms integrated into the fundamental structure of the computer [20], one would instead split the computational horsepower up into multiple smaller decentralized pieces.[21, 22] Each of these pieces would then represent a single quantum node, that is able to create entanglement between a stationary atom and a mediating qubit, transfer the state of the mediating qubit onto the stationary one, and also process information through the use of quantum logic gates. By linking many of these quantum nodes one aims to overcome size-scaling and error-correction problems that would limit the size of machines for quantum processing.[22, 23]. This network of nodes would then appropriately be named a *Quantum Internet*. One way of realizing such a quantum internet is via the strong coupling interaction of single photons and atoms in the setting of cavity quantum electrodynamics (*cQED*). Here, each quantum node would store a set number of stationary atoms in between two highly reflective mirrors, which are able to be individually controlled, and interlinked with one another through another set of quantum channels, realized by photons that travel in between the nodes through optical fibers.

One of these cavity systems is the *CavityX* system at the Max Planck Institute for Quantum Optics which features a ^{87}Rb atom trapped between two crossed fiber optic cavities, hence the letter X in the name. In the past this setup has been used to achieve atom-photon entanglement with a fidelity of 87% [24], successfully transfer the information from a photonic qubit onto an atomic qubit with a fidelity of 94.7%[25], and recently realize atom-atom entanglement between two nodes connected via a fiber going through the Munich city center all the way out to the research campus in Garching while still achieving a fidelity of include distance, number and citation here. However as mentioned before, in order to realize a full quantum internet, the last milestone is the realization of the processing nature of the quantum node part mentioned in the definition of the quantum internet by Jeff Kimble [22].

This is where this thesis comes into play. The goal of this thesis is to realize an atom-photon reflection gate using one of the two fiber cavities in the CavityX setup, that one being birefringent, while preparing the atom in either of the two clock states of the D_2-line ground states, thus minimizing the influence of external factors like magnetic field fluctuations. Using a setup inspired by Duan and Kimble [21] and storing the information in the polarization of a photonic qubit has the advantage of allowing for controlled gate operations between a flying photon and a resonantly coupled atom trapped inside of the cavity. With the realization of a quantum gate between a stationary atom and a flying photonic qubit, one would be able to reuse this setup to build further identical quantum nodes and thus completing the final requirement for a fully functional quantum network node, ready for integration into a Quantum Internet.

The structure of this thesis is as follows. Chapter 2 introduces the theoretical framework of cavity quantum electrodynamics, including the effects of cavity birefringence and the mechanism of atom-state-dependent cavity reflection that underpins the gate operation. Chapter 3 describes the experimental platform, covering the cavity system, the trapping and positioning of a single ^{87}Rb atom, and the preparation of the atomic qubit in the clock states. Chapter 4 details the photonic qubit path, addressing polarization-dependent losses and the input-output optical setup. Chapter 5 presents the main results: the characterization of the atom-photon gate performance through gate truth tables and the generation of Bell states. Finally, Chapter 6 provides a summary and outlook.

Chapter 2

Theory

This chapter introduces the theoretical framework needed to understand the key points of this thesis. First a general introduction into cavity quantum electrodynamics will be given, alongside the intricacies in the setup that will modify the general theory. Afterwards the reflection from an atom-cavity system will be explored, and finally how it can be used to construct an atom photon gate.

2.1 Cavity Quantum Electrodynamics

The field of *Cavity Quantum Electrodynamics* explains the interaction between a resonator (in this case a Fabry-Perot cavity), and a piece of matter with which the light inside the resonator interacts (here: a ^{87}Rb atom). By placing the atom between two highly-reflective mirrors forming the optical cavity, one increases the number of times a photon that is inside a cavity bounces back and forth between the two mirrors, enhancing the interaction cross-section compared to free space. The strength of this enhancement is captured by the single-atom cooperativity $C = g^2/(2\kappa\gamma)$, which must satisfy $C > 1$ for deterministic light-matter interaction.[26]
The theoretical model which explains this interaction in a fully quantum mechanical picture is called the *Jaynes-Cummings Model* [27] and was named after Edwin Jaynes and Fred Cummings who developed it in 1963. To introduce this model, we consider a three-level emitter (i.e. the atom), which is interacting with a single cavity optical mode.
The *Jaynes-Cummings* (JC) Model describes the coherent exchange of energy between a quantized cavity field and a two-level transition of the atom. The cavity supports a single optical mode at frequency ω_c. The atom, which is also located in the cavity, supports a transition frequency close to ω_c, from a coupled state $|c\rangle$ to an excited state $|e\rangle$ at a frequency ω_a. This transition is treated in the dipole approximation. In order to then complete the three-level-emitter picture, we include an uncoupled state $|u\rangle$ with which the photon does not interact due to this state not being resonantly coupled to the cavity light mode. This uncoupled state $|u\rangle$ is what enables the conditional atom-photon gate operation.
Lastly, any real atom-cavity system, additionally suffers from losses, primarily the spontaneous decay of the atomic excitation at a rate of 2γ and the cavity field decay at a rate κ.
The full Hamiltonian which describes interaction between the atom and the cavity thus reads: [27]

$$\mathcal{H}_{\mathcal{JC}} = \underbrace{\hbar\omega_c \hat{a}^\dagger \hat{a}}_{\widehat{=}\text{cavity field}} + \underbrace{\hbar\omega_a \hat{\sigma}_{ce}^\dagger \hat{\sigma}_{ce}}_{\widehat{=}\text{atom}} + \underbrace{\hbar g(\hat{a}\hat{\sigma}_{ce}^\dagger + \hat{a}^\dagger \hat{\sigma}_{ce})}_{\widehat{=}\text{light-atom interaction}} \tag{2.1}$$

The three terms in (2.1) describe the energy of the cavity field, the energy of a bare atom, and the coherent coupling between the two governed by a coupling strength g.
The operators $\hat{a}^\dagger$ and $\hat{a}$ are the creation and annihilation operators of the cavity light modes, whereas $\hat{\sigma}_{ce}^\dagger = |e\rangle\langle c|$ and $\hat{\sigma}_{ce} = |c\rangle\langle e|$ are the atomic raising and lowering operators, respectively.

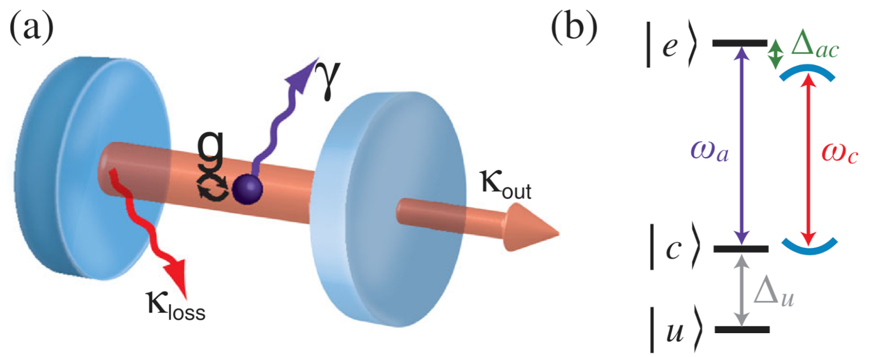

Figure 2.1: (a) Basic physical situation considered in cQED. A three-level emitter, e.g., a single atom, is located in an optical cavvity, schmeatically represented by two facing mirrors. The atom exchanges energy with the optical field at a rate $2g$. The decay rate of photons out of the resonator is $2\kappa = 2\kappa_{\text{out}} + 2\kappa_{\text{loss}}$. The atomic polarization dewcay rate is γ (b) Atomic energy level scheme. The transition frequency between the coupled ground state $|c\rangle$ and the excited state $|e\rangle$ is ω_a. The cavity resonance frequency is ω_c. Atom and cavity are detuned by $\Delta_{ac} = \omega_a - \omega_c$. The third, uncoupled level $|u\rangle$ is detuned from $|c\rangle$ by Δ_u.[26]

The form of (2.1) assumes the so called *Rotating-Wave Approximation* in which fast oscillating terms at optical frequencies can be neglected.
Solving for the eigenenergies of the system one is left with:

$$E_0 = 0 \tag{2.2}$$

$$E_\pm(n) = n\hbar\omega_c + \hbar\frac{\Delta_{ac}}{2} \pm \frac{\hbar}{2}\sqrt{\Omega_n^2 + \Delta_{ac}^2}, \quad n \in \mathbb{N} \tag{2.3}$$

In (2.3), n is the number of excitation quanta in the system, with $\Omega_n = 2g\sqrt{n}$ being the corresponding n-quanta Rabi frequency. The term Δ_{ac} describes the detuning of the cavity resonance frequency from the atomic transition. The main take-away from (2.3) is that on resonance ($\omega_a = \omega_c$), the eigenenergies of the system are split by $2\hbar g\sqrt{n}$, giving rise to the so-called *Vacuum Rabi Splitting*.

2.1.1 Cavity Birefringence

Equation (2.1) only covers the case for having a single cavity light-mode. The setup used in this thesis, however, features a birefringent cavity. The birefringence is created by physically changing the geometric shape of the cavity mirror during production of the fiber ends.[28] This forces a change in the optical path lengths between different polarizations, leading to a frequency splitting $\Delta\nu$ proportional to the ellipticity ϵ of the mirror surfaces:[29]

$$\Delta\nu = \frac{\nu_{FSR}}{2\pi k}\frac{\epsilon^2}{R_2}. \tag{2.4}$$

The property ν_{FSR} in (2.4) describes the free spectral range, $k = \frac{2\pi}{\lambda}$ the photon wavenumber, and R_2 the radius of curvature of the cavity mirror.
This means that we need to consider a system with two different light modes for the JC Hamiltonian, which in this case are the π and the V modes. However only the π mode is resonantly coupled to the $|c\rangle \leftrightarrow |e\rangle$ transition, whereas the V mode is detuned by $\Delta_{\text{biref}} = 2\pi \cdot 500\text{MHz}$. In the context

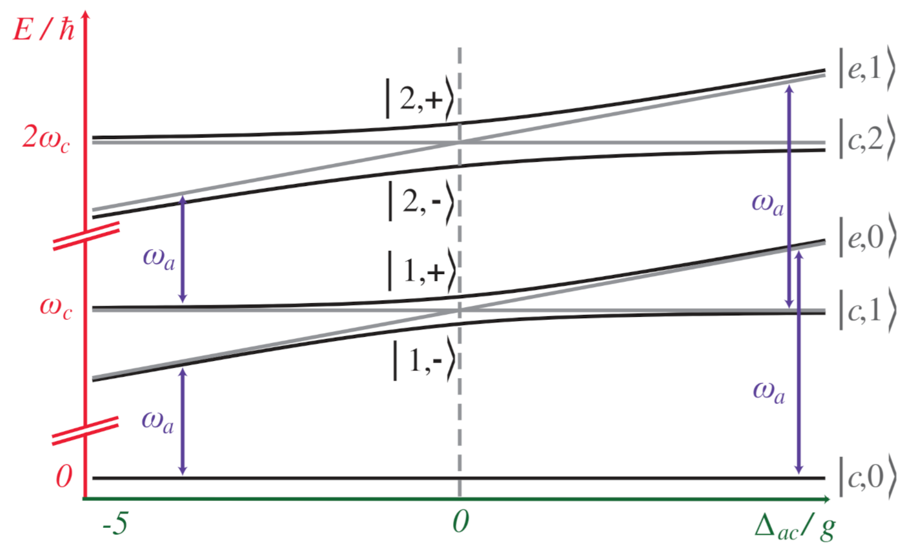

Figure 2.2: The Jaynes-Cummings ladder of the dressed energy eigentstates $|N, \pm\rangle$ of a strongly coupled atom-cavity system, where the cavity frequency is kept fixed and the emiter frequency is scanned over the resonance. Compared to the uncoupled situation (gray), the coupled states (black) exhibit pronounced avoided level crossings. At zero detuning, the levels are separated by $2g\sqrt{N}$.[26]

of this thesis $\Delta_{\text{biref}} \gg \kappa, g, \gamma$. This effectively makes the V-mode uncoupled, and can be neglected in the analytical treatment. It is, however, still included in the Hamiltonian for use in numerical simulations where both modes are driven explicitly. This leads to the extended Hamiltonian:

$$H_{JC} = \hbar\omega_{c,\pi}\hat{a}_\pi^\dagger\hat{a}_\pi + \hbar(\omega_{c,\pi} + \Delta_{\text{biref}})\hat{a}_V^\dagger\hat{a}_V + \hbar\omega_a\hat{\sigma}_{ce}^\dagger\hat{\sigma}_{ce} + \hbar g(\hat{a}_\pi\hat{\sigma}_{ce}^\dagger + \hat{a}_\pi^\dagger\hat{\sigma}_{ce} + \hat{a}_V\hat{\sigma}_{ce}^\dagger + \hat{a}_V^\dagger\hat{\sigma}_{ce}) \quad (2.5)$$

2.1.2 External Drive

At the heart of the atom-photon gate lies the reflection of the incoming photon off the cavity system. In order to describe this in a quantum mechanical way, one needs to include a driving term in the Hamiltonian with the form of $H_{D,S} = i\hbar(\epsilon e^{-i\omega_d t} + \epsilon^* e^{i\omega_d t})(\hat{a} + \hat{a}^\dagger)$. This Hamiltonian $H_{D,S}$ is defined in the Schrödinger picture, and has a complex drive amplitude ϵ and driving field frequency ω_d. After applying the rotating wave approximation (RWA), which neglects the fast oscillating terms in $H_{D,S}$, and changing into the rotating wave frame to remove the explicit time dependence, one gets the following expression:

$$H_D = \mathrm{i}\hbar\left(\hat{a}_s\epsilon^* + \hat{a}_s^\dagger\epsilon\right) \quad (2.6)$$

where $\hat{a}_s$ denotes the annihilation operator of the cavity mode matching the input photon polarization (s = π or V).
Adding (2.6) to (2.5), one gets the following full Hamiltonian:

$$\begin{aligned} H_{JC} = &\hbar\Delta_{c,\pi}\hat{a}_\pi^\dagger\hat{a}_\pi + \hbar\left(\Delta_{c,\pi} + \Delta_{\text{biref}}\right)\hat{a}_V^\dagger\hat{a}_V + \hbar\Delta_a\hat{\sigma}_{ce}^\dagger\hat{\sigma}_{ce} \\ &+ \hbar g\left(\hat{a}_\pi\hat{\sigma}_{ce}^\dagger + \hat{a}_\pi^\dagger\hat{\sigma}_{ce} + \hat{a}_V\hat{\sigma}_{ce}^\dagger + \hat{a}_V^\dagger\hat{\sigma}_{ce}\right) \\ &+ \mathrm{i}\hbar\left(\hat{a}_s\epsilon^* + \hat{a}_s^\dagger\epsilon\right) \end{aligned} \quad (2.7)$$

This then gives a definition of the JC Hamiltonian with respect to an external driving field, and based on the detunings of the driving field with respect to the cavity resonance frequency $\Delta_{c,\pi} = \omega_{c,\pi} - \omega_d$ and the atomic transition frequency $\Delta_a = \omega_a - \omega_d$, respectively.

2.2 Reflection from an Atom-Cavity System

For quantum information processing, it is essential that the photonic qubit is preserved upon reflection from the cavity. This requires that light entering through the first mirror called the outcoupling (OC) mirror also exits through the same mirror, rather than being lost through the opposing highly reflective (HR) mirror. The HR mirror on the other hand must have maximum reflection, making the cavity one sided, so the light exits predominantly through the OC mirror. The total losses can thus be divided into the following parts:

$$\kappa = \kappa_{oc} + \kappa_{HR} + \kappa_{paras} \tag{2.8}$$

In (2.8), κ_{oc} and κ_{HR} describes the rate at which light couples into and out of the cavity through the OC and HR mirror respectively, whereas κ_{paras} describes the rate of losses due to absorption at the mirror surfaces. In order for the light to enter and leave through the OC mirror, one needs to ensure that $\kappa_{oc} \gg \kappa_{HR}, \kappa_{paras}$. The escape probability of the photon is then defined by the outcoupling ratio $\frac{\kappa_{oc}}{\kappa}$.

Another important quantity is the matching of the spatial modes of the light when coupling into the cavity. As the incoming light gets coupled into the cavity, a mismatch between the spatial modes in transmission and reflection can occur. This lowers the efficiency at which the light can couple to the cavity itself. The mode matching is then given by the overlap integral of the mode functions for the light one is investigating. In this case it is important to quantify the mode matching between the reflected light at the first cavity and the incoming light (μ_{fr}) as well as the one for the incoming light and the optical mode inside of the cavity (μ_{fc}).

The field incident on the cavity is treated as an input field. This then can be linked to an output field via input-output formalism. When driving the system only from one side, as it is the case here, the relation between input and output becomes:[30]

$$a_{out}(t) = \mu_{fr} a_{in}(t) + \mu_{fc}\sqrt{\kappa_{oc}}\langle a\rangle(t) \tag{2.9}$$

Here (2.9) was adjusted to also consider the mode matching parameters. Solving Langevin equations for weak coherent driving and photonic wave envelopes which vary only on a time scale longer than κ, one can write down the analytical representation of the reflection amplitude[31, 32, 33]:

$$r(\omega) = \mu_{fr} - \mu_{fc}^2 \frac{2\kappa_{oc}(i\Delta_a + \gamma)}{(i\Delta_{c,\pi} + \kappa)(i\Delta_a + \gamma) + g^2} \tag{2.10}$$

The reflection amplitude in (2.10) depends on the detuning $\Delta_{c,\pi} = \omega - \omega_{c,\pi}$ between the driving laser frequency and the cavity resonance frequency $\omega_{c,\pi}$ (for the π mode) and on the detuning relative to the atomic transition frequency $\Delta_a = \omega - \omega_a$.

Include reflection spectrum for different detunings and also phase across different detunings

2.3 Atom-State-Dependent Cavity Reflection

Looking at (2.10) one can see that the behavior of the reflection amplitude at zero detuning ($\Delta_a = \Delta_{c,\pi} = 0$) is:

$$r(\Delta = 0) = \mu_{fr} - \mu_{fc}^2 \frac{2\kappa_{oc}\gamma}{\kappa\gamma + g^2} \tag{2.11}$$

In the event of an uncoupled atom-cavity system (effectively $g = 0$), which would be achieved by preparing the atom in the $|u\rangle$ state, the reflection coefficient $r < 0$. This would imply a relative phase shift of π between the reflected and incoming light field ($a_{out}(t) \approx -a_{in}(t)$). Preparing the atom in the coupled state $|c\rangle$ however, the Vacuum Rabi Splitting shifts the cavity resonances to

$\omega_c \pm g$, so the cavity appears off-resonant for a photon at ω_c. The photon is therefore reflected from the OC mirror without entering the cavity, acquiring no phase shift, which is mirrored by a reflection coefficient of $r > 0$ ($a_{out}(t) \approx a_{in}(t)$). A V polarized photon, being far detuned from the cavity resonance by Δ_{biref}, is reflected without entering the cavity and acquires no phase shift, regardless of the atomic state. This state-dependent phase shift of the output field depending on the state of the atom, enables the realisation of a quantum atom-photon gate with the atom being the controlling qubit and the photon being the target qubit.[21]
Adopting standard quantum computing notation, the uncoupled state of the atom is the $|0\rangle$ state, and the coupled one the $|1\rangle$ state. The photon, either π-polarized or V-polarized, is represented as the input states $|\pi\rangle$ and $|V\rangle$. Furthermore, for all input states, the input drive is resonant with the π mode of the birefringent cavity.
For an input state $|0, \pi\rangle$, meaning the atom is prepared in the uncoupled state $|0\rangle$ and a π-polarized photon is sent in, the photon will be able to enter the cavity and (ideally) exit through the OC mirror, with a π phase-shift. For the input state $|1, \pi\rangle$, the Vacuum Rabi splitting shifts the cavity resonance away from the photon frequency, causing it to be reflected without a phase shift. When sending a $|V\rangle$ photon, it will get reflected off of the OC mirror regardless, due to the birefringence of the cavity. This results in the following reflections:

$$
\begin{aligned}
|0, \pi\rangle &\rightarrow - |0, \pi\rangle \\
|0, V\rangle &\rightarrow |0, V\rangle \\
|1, \pi\rangle &\rightarrow |1, \pi\rangle \\
|1, V\rangle &\rightarrow |1, V\rangle
\end{aligned}
\tag{2.12}
$$

This output from the atom-photon reflection gives us a *Controlled Phase Flip* (CPF) gate. This conditional phase shift allows the construction of a universal quantum gate that can be transformed into any two-qubit gate using rotations of the individual qubits, which in the case of the photon can be done via retardation waveplates.
Rotating the photonic basis states to:

$$
\begin{aligned}
|+\rangle &= \frac{|V\rangle + |\pi\rangle}{\sqrt{2}} \\
|-\rangle &= \frac{|V\rangle - |\pi\rangle}{\sqrt{2}}
\end{aligned}
\tag{2.13}
$$

and keeping the atom either in the coupled ($|1\rangle$) or uncoupled case ($|0\rangle$), results in the following reflection:

$$
\begin{aligned}
|0, +\rangle &\rightarrow |0, -\rangle \\
|0, -\rangle &\rightarrow |0, +\rangle \\
|1, +\rangle &\rightarrow |1, +\rangle \\
|1, -\rangle &\rightarrow |1, -\rangle
\end{aligned}
\tag{2.14}
$$

In the new photonic basis $\{|+\rangle, |-\rangle\}$, the CPF gate is equivalent to a *Controlled-NOT* (CNOT) gate, with the atom as the controlling qubit and the photon polarization as the target qubit. The photon polarization is flipped ($|+\rangle \leftrightarrow |-\rangle$) when the atom is in $|0\rangle$, and left unchanged when the atom is in $|1\rangle$.

Chapter 3

Atom-Cavity platform

The system uses a single ^{87}Rb coupled to a birefringent Fabry-Perot cavity inside an ultra-high vacuum (UHV) chamber. The smaller optical beam path used for the realization of the atom-photon gate is part of a much larger setup. The full setup, named *CavityX*, consists of two Fabry-Perot cavities arranged in an X formation. For the atom-photon gate, parts of the setup like the magneto-optical trap, the 850 nm red-detuned trap laser, and the microwave antennas integrated into the UHV chamber are crucial. Other components, most notably the second cavity, are not within the scope of this thesis, but will be discussed in the outlook.

3.1 Cavity system

At the heart of the CavityX system are the two crossed fiber cavities which are named the "short" cavity (SC) and the "long" cavity (LC), which differ in length ($79\,\mu$m vs. $161\,\mu$m), and in the fact that the short cavity is birefringent. Each cavity fiber is held in place by an in-house built holder shown in figure 3.1.
The incoming light is coupled in and out via the single-mode (SM) fiber for either of the cavities. In order to ensure that most of the coupled light exits through the SM fiber, different mirror coatings are used for the fiber surfaces. The multi-mode (MM) fiber surfaces are coated with a highly reflected dielectric coating, while the opposing SM fiber has a dielectric coating with a lower reflectivity (10 ppm transmission vs. 340 ppm transmission). This asymmetric design ensures the cavity is single-sided, causing most of the coupled-in light to then exit the cavity through the same fiber through which it entered the system.[34]
The cavity fibers are fabricated using a CO_2 laser machining technique by Manuel Brekenfeld in [28]. Changing the beam profile of the CO_2 laser allows for the fabrication of either spherical or elliptical surfaces, as is the case for the short cavity. This leads to a cavity mode splitting, i.e., cavity birefringence, as shown in figure 3.2

3.2 Trapping of a Single Atom

3.2.1 Magneto-optical trap

In order to realize an atom-photon quantum gate, one is required to have an atom trapped inside the cavity with which the photon can then interact. Therefore one needs to properly position the atom and keep it trapped at the position where it most efficiently couples to the cavity to ensure the most efficient gate operation.

The starting point for this is the trapping of an atomic cloud via a magneto-optical trap (MOT). Once the atomic cloud is trapped, one turns off the MOT, causing the atoms to fall down due to

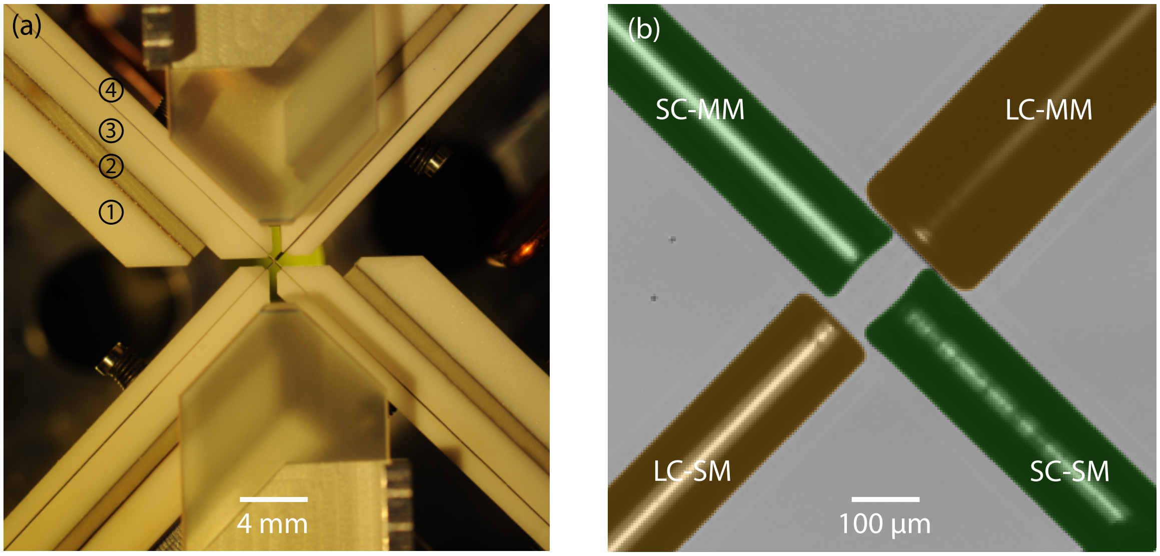

Figure 3.1: vertical top view of the construction holding the crossed fiber cavities in place. The holder of each fiber consists of 4 elements, each of which is labeled in the picture: ① Base element for mounting the fiber holder to downstream components; ② Shear piezo actuator for active stabilization of the resonator length; ③ Mount in which a V-groove is fabricated - where the fiber is positioned and mechanically clamped; ④ Lid to clamp the fiber within the V-groove mount. A zoom of the center of the picture is shown in (b). The orange-colored cavity has a length of 161 m and it is composed of a single-mode fiber (LC-SM) with a low reflective mirror coating and a multi-mode fiber (LC-MM) with a highly reflective mirror coating. The green-colored cavity is 75.7 m long and is composed of a single- (SC-SM) and multi- (SC-MM) mode fiber with reflective coatings analogous to LC-SM and LC-MM, respectively. This photo has been colored in post-production.[34]

the gravitational force, from which single atoms can then be loaded into the cavity's center. The concepts of a MOT are explained in [35].
In order to trap atomic clouds via a MOT, one needs to generate a rubidium vapor inside the UHV chamber. This is done via light-induced atom desorption (LIAD), in which atoms are dispensed from a rubidium dispenser that is placed inside of the vacuum chamber. The light source is a UV diode ILH-XQ01-S410 with an emitting wavelength of 410nm, which is pulsed for the length of time needed to load the MOT.

The MOT loading procedure can be divided into two phases: First comes the loading phase in which the MOT coils, the MOT beams (a cooling laser and a repumper laser), and the UV light are switched on simultaneously. This then creates the atomic cloud and cools it down to near the Doppler temperature of 146 μK for the ^{87}Rb D_2-line. [36]
Next an optical molasses is applied, which aims at bringing the atom to a temperature much below the Doppler limit. Here the detuning of the MOT beams is increased, while the optical power is decreased.
At the end of the MOT cycle, the MOT coils, beams and the UV light are switched off.

3.2.2 Trapping and Positioning of Single Atom

At the end of the MOT loading phase, the ultra-cold atomic cloud then sits approximately 10 mm above the cavity plane. Turning off the MOT causes the atoms to fall towards the cavity due to gravity, until they reach the cavities' crossing point, at which point cooling and trapping laser beams are switched on.
The cooling and repumping laser are shone diagonally onto the crossing point in a counter-propagating

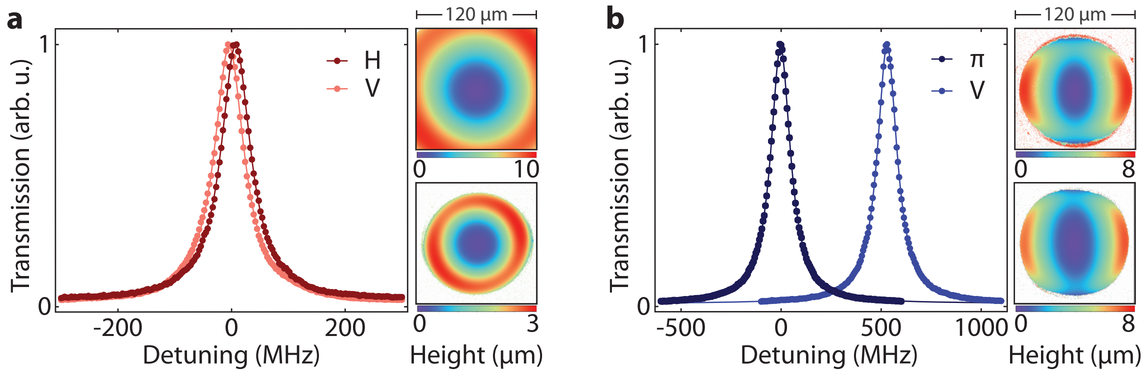

Figure 3.2: Transmission spectra were recorded for both the long (a) and short (b) cavity. For the long cavity, the probe field was polarized to match the cavity eigenmodes, revealing a small residual polarization mode splitting. Contour images of the mirrors for the MM and SM fibers are shown alongside the spectrum. In contrast, the short cavity exhibited a large polarization mode splitting, a result of the strong ellipticity of its mirror surfaces. The error bars are smaller than the data point markers.[34]

configuration. Using orthogonal polarizations in a lin⊥lin configuration for the counter-propagating lasers, the atoms are cooled to sub-Doppler temperatures via Sisyphus-cooling.
Once the atom is at the center of the cavity, two lasers are responsible for keeping it trapped there: The first one is the intracavity laser of the short cavity which has a wavelength of 772 nm, and is blue-detuned from the D_2-line. The laser self-interferes in the cavity, forming a standing wave, thus confining the atom at the nodes, i.e., interference minima. Stabilizing the cavity length is achieved by first locking the intracavity laser using the PDH-lock technique and then locking the laser frequency to an optical frequency comb, which serves as a stable frequency reference. For the atom-photon gate the frequency of the laser was chosen so that the cavity is resonant to the D_2-line at 780 nm.

The second laser is the 852 nm vertical trap laser, that's shone in from the above the cavities' crossing point. The laser is linearly polarized along the long cavity axis, and focussed to a diameter of about 20 μm. It is then transmitted through the chamber and collimated to a mode diameter of 1cm. Lastly it gets transmitted through a dichroic element, a set of retardation waveplates, a piece of silica glass mounted to a galvanometer scanner (galvo), to then get retroreflected in a cat-eye configuration. Due to the laser being far detuned from the D_1 and D_2 lines of the atom, it forms a trapping standing wave, where the atoms are confined in the antinodes, i.e., interference maxima.

Include image from gianvito's thesis about setup

In order to achieve maximum cavity coupling, the galvo scans its amplitude, rotating the silica glass and thereby modifying the beam's angle of refraction, which shifts the standing-wave antinodes vertically. While this is happening, the fluorescence counts coming from the atom during the cooling phase are collected via the short cavity and detected with superconducting nanowire single-photon detectors (SNSPDs). The fluorescence counts are monitored in real time by an FPGA, which also controls the galvo scan amplitude. One can use this determine the height and thus the maximum coupling position of the atom from the fluorescence counts. In order to ensure that only a single atom was trapped instead of two weakly coupled atoms, the FPGA sequence is as follows:

1. The FPGA starts scanning the voltage of the galvo while monitoring the amount of fluorescence counts detected in a 25 ms window coming from the atom.

2. If the fluorescence counts go above a certain threshold (noise-threshold), this means that an atom that is weakly coupled to the cavity is emitting fluorescence counts into the cavity.

Table 3.1: Cavity parameters.

Parameter	Long Cavity	Short Cavity
Radius of curvature of OC_1	340 μm	115 μm and 283 μm
Radius of curvature of HR_2	170 μm	96 μm and 167 μm
Cavity length	161 μm	75.7 μm
Free spectral range	0.93061 THz	1.98043 THz
Waist radius	3.5 μm	3.5 μm and 4.4 μm
Waist radius pos. from OC	7.7 μm	25.8 μm and 22.7 μm
Mode waist at OC	3.6 μm	3.9 μm and 4.6 μm
Mode waist at HR	11.3 μm	5.9 μm and 5.2 μm
Mode waist at cavity center	6.2 μm	3.5 μm and 4.4 μm
Mode volume ($\lambda = 780\,\text{nm}$)	$10200 \cdot \lambda^3$	$1900 \cdot \lambda^3$
Mode matching		write down actual μ_{fc} here
$g(1' \leftrightarrow 2)/(2\pi)$ at cavity center		write down actual g here
Transmission through OC	340 ppm	340 ppm
Transmission through HR	10 ppm	10 ppm
Parasitic losses	80 ppm	50 ppm
Total losses	430 ppm	400 ppm
$\kappa/(2\pi)$		write down κ here
Linewidth		write down LW here
Finesse	14089	16531
Outcoupling ratio		write down outcoupling ratio here
Eigenmodes freq. splitting	16 MHz	505.0(5) MHz

Table 3.2: write down table title and also notes regarding the subscripts in there

3. The FPGA then continues increasing the voltage, keeping track of the voltage (i.e., position) at which the maximum fluorescence counts were reached. In the meantime the clicks at each galvo step are summed into a single 32-bit register. This effectively represents an integral over the fluorescence counts as a function of the galvo amplitude curve.

4. Once the fluorescence counts drop below the (noise-threshold), meaning that the atom has been moved out of the cavity center, the integral of the fluorescence counts is evaluated. If the integral is within a certain range, the FPGA will set the voltage at which maximum fluorescence counts were detected. If the integral is out of range (too low: atom was badly coupled anyway; too high: multiple atoms were in one node), the galvo keeps scanning until it either finds a new atom or reaches the maximum amplitude, causing the MOT and single-atom positioning loop to start again.

A different atom-positioning algorithm was employed in the past (e.g. the one used in Gianvito Chiarella's thesis [34]), where the FPGA only scanned up to the point where the maximum fluorescence counts were reached. This past algorithm could load two atoms which were badly coupled to the cavity since a doubly-loaded node was indistinguishable from a well-coupled single atom. By completing the scan over the entire fluorescence counts curve, one is able to properly ensure single atoms are being trapped. Mention something like the g2 or some metric that shows how this algorithm performs and also how it improved over the last algorithm, also do same measurement as gianvito in his thesis to showcase the loading algorithm cuz coool

3.3 Atomic Qubit Preparation

For the atom-photon gate, the single atoms must be prepared in the two $m_F = 0$ ground states of the D_2-line, those being $5S_{1/2}, F = 1, m_F = 0$ for the uncoupled state, and $5S_{1/2}, F = 2, m_F = 0$ for the coupled state.

Table 3.3: Parameters used for the MOT loading phase.

MOT coil current	23.5 A
UV diode current	140 mA
MOT beams' diameter	12 mm
Loading phase	
Duration	check time for MOT
Cooling laser detuning from $5S_{1/2}, F = 2 \leftrightarrow 5P_{3/2}, F' = 3$	check detuning again
Repumping laser detuning from $5S_{1/2}, F = 1 \leftrightarrow 5P_{3/2}, F' = 2$	0 MHz
Cooling laser power	20 mW
Repumping laser power	250 μW
Molasses phase	
Duration	1 ms
Cooling laser detuning from $5S_{1/2}, F = 2 \leftrightarrow 5P_{3/2}, F' = 3$	−31 MHz
Repumping laser detuning from $5S_{1/2}, F = 1 \leftrightarrow 5P_{3/2}, F' = 2$	−20 MHz
Cooling laser ramp-down power	1.7 mW
Repumping laser power	250 μW

Parameters used for the MOT loading phase. The UV diode and the MOT coils are permanently on the whole time. The cooling and repumping laser power is measured behind a polarizing beam splitter, which separates the horizontal MOT beam from the top-diagonal ones.

The first step in achieving this is to optically pump (OP) the atom into the desired state. This is done by repeatedly exciting the atom from a level one wishes to depopulate to an excited state, from which it will then decay into various ground states, determined by the branching ratios of the respective transition. In order to then populate the desired state, one uses a frequency for the pump laser, where all ground states one wants to depopulate are coupled to the excited state, except for the desired one, gradually populating the target state over time.
In this work, one first pumps the atom into the $F = 1, m_F = 0$ state. This is done by using two laser beams, one that is resonant with the $F = 2 \rightarrow F' = 2$ transition of the D_2-line of ^{87}Rb, which will lead to the depopulation of the $F = 2$ manifold and population of the $F = 1$ manifold. In order to then populate only the $F = 1, m_F = 0$ state, one applies another beam, resonant to the $F = 1 \rightarrow F' = 1$ transition of the D_2-line. Furthermore the beam must be π-polarized, since the $F = 1, m_F = 0 \rightarrow F' = 1, m_F = 0$ transition is forbidden.[36]. The transition scheme can be seen in figure 3.3.
Using this technique one is able to populate the $F = 1, m_F = 0$ level with a probability of $\geq 97\,\%$.[34]

For the atom-photon gate, one also needs to be able to prepare the $5S_{1/2}, F = 2, m_F = 0$ for the coupled state. This is achieved by first preparing the atom in the already prepared $F = 1, m_F = 0$ state and using a set of microwave antennas integrated into the UHV chamber, to drive the $|F = 1, m_F = 0\rangle \leftrightarrow |F = 2, m_F = 0\rangle$ clock transition.
Finding the transition frequency is done by performing a microwave spectroscopy measurement. The atom is pumped into the $F = 1, m_F = 0$ via the previously discussed technique. By scanning the microwave frequency at fixed phase, one measures the F=2 population via cavity-enhanced fluorescence state detection (SD) with a detection window of 7.5 μs.
The resulting MW spectrum shown in **??** then shows three peaks, which correspond to the $F = 1, m_F = 0 \leftrightarrow F = 2, m_F = 0$ and $F = 1, m_F = 0 \leftrightarrow F = 2, m_F = \pm 1$ transitions. From this one can infer that the optical pumping into the $F = 1, m_F = 0$ is done with an efficiency of 97(1)%.[34]
show MW spectroscopy regarding the transition from $F = 1 m_F = 0$ to the other states
Once the frequency for the $F = 1, m_F = 0 \leftrightarrow F = 2, m_F = 0$ transition is known, one can coherently manipulate the atomic population in the ground state. By tuning the MW on resonance, one can drive Rabi oscillations [37] between two ground states, where the population evolution in the Zeeman

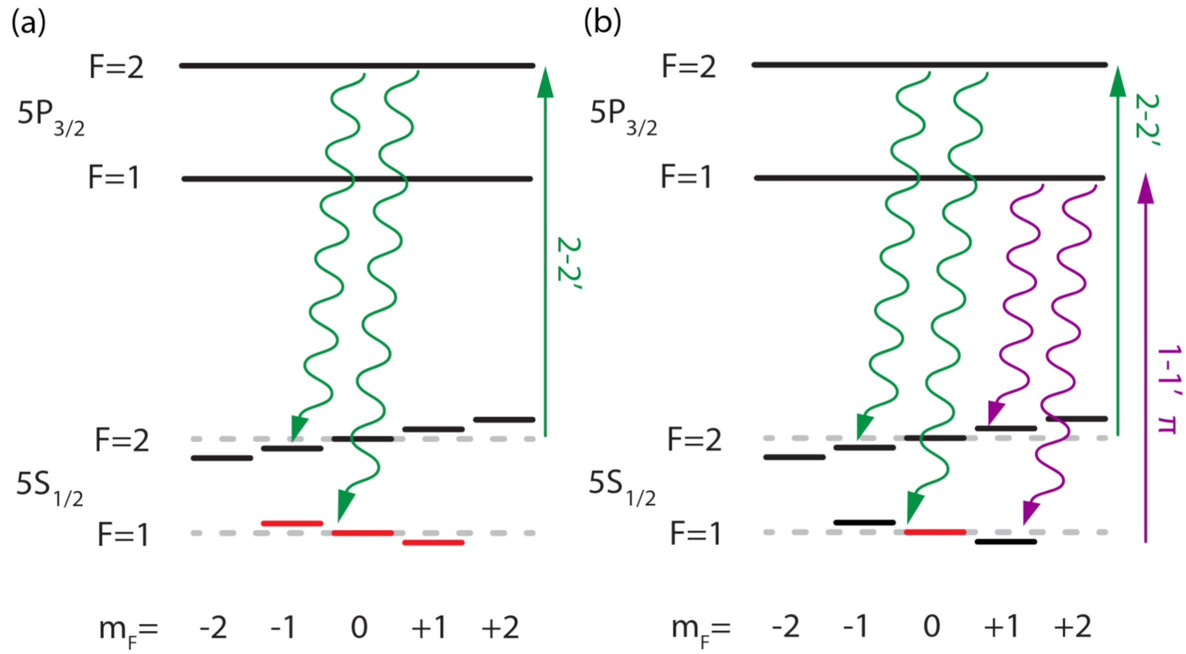

Figure 3.3: Schemes utilized for OP in $F = 1$ (a), and $F = 1, m_F = 0$ (b).Each sketch shows the used lasers (straight arrows), the atomic decay (wavy arrows), and the targeted state (red lines).[34]

state of $F = 2$ can be modeled as:[38]

$$P_2 = P_1 \cdot \frac{1}{2}(1 - \cos(2\pi\Omega t) \cdot e^{-\chi t}) \tag{3.1}$$

where P_1 is the initial population after optical pumping in the state $F = 1, m_F = 0$, Ω is the previously determined Rabi frequency, and χ is the damping rate. The result is a combination of a sinusoidal term which describes the coherent state transfer between the two levels, and an exponential factor representing the dissipative mechanism which damps the oscillations.
By scanning the length of the π pulse at a constant MW frequency, one can determine the ideal time it takes for a state to coherently transfer from $F = 1, m_F = 0$ to $F = 2, m_F = 0$. Figure **??** shows the resulting Rabi oscillations for an atom prepared in $F = 1, m_F = 0$ and transferred to the $F = 2, m_F = 0$ state. Fitting equation (3.1): write down fit params

Chapter 4

Photonic Qubit Path

4.1 Polarization-Dependent Losses

Looking at equation (2.10), it is apparent that the reflection amplitude for a photon that gets reflected from an atom-cavity system, is highly dependent on specific system parameters. In this case those are the coupling strength g, the decay rate of the photons κ and the outcoupling ration κ_{OC}, aswell as the two mode-matching parameters μ_{fr} and μ_{fc}. Furthermore, one needs to also take into account the birefringence of the cavity itself, thus making the parameter Δ_{biref} also very important (here: $\Delta_{\text{biref}} = 2\pi \cdot 500\text{MHz}$). Looking at the reflection scheme for (2.14) in the basis defined in (2.13), it becomes apparent that in order to achieve the highest possible overlap with an ideal CNOT gate, the reflection in both the π and the V mode, must be (ideally) the same. However, using the parameters measured in section 5.1, this is not the case.

4.2 Inducing Losses via a Brewster Window Array

Brewster windows are a piece of uncoated UV fused silica, when placed at the so called brewster angle, that the p-polarized portion of the incomming light gets transmitted without any losses, while the s-polarized portion of the light gets partially reflected.
Using the system parameters and calculating the reflection amplitude via (2.10) it was apparent that the reflection from the atom-cavity system, introduces extra polarization dependent losses in the π mode as opposed to the V mode. Although the effect on the reflection in the CPF basis wouldn't be of any concern, it would lead to a reduced fidelity when measuring in the CNOT basis, as there the two polarization modes would need to have the same amplitude in reflection in order to show the expected flip from one state to the other.
In order to combat this and keep keep the fidelity high, an array of brewster plates were installed on a custom machined mount, which place the brewster plates at the brewster angle relative to the incoming light, so that the V polarized portion of the incoming light gets attenuated prior to interacting with the atom-cavity system. This ensures that in reflection, both polarization modes have the same reflection amplitude, allowing for the observation of the flip on the photonic qubit.
The brewster angle can be simply calculated via:[39]

$$\tan\theta_B = n_t/n_i \tag{4.1}$$

$$n_t^2 - 1 = \frac{0.6961663\lambda^2}{\lambda^2 - 0.0684043^2} + \frac{0.4079426\lambda^2}{\lambda^2 - 0.1162414^2} + \frac{0.8974794\lambda^2}{\lambda^2 - 9.896161^2} \tag{4.2}$$

In (4.2) n_t is the optical density of the brewster window, and n_i would be the optical density from which the incoming light is coming from (here: air $n_i \sim 1.004$)
The custom brewster window mount as mentioned before has the main functionality of placing the elements in an array at the brewster angle θ_b relative to the incoming light. It can house up to 12

brewster windows when using the screw whole in the middle or 8 brewster windows, which allows the mount to be placed on two seperate posts, making the overall mount more stable. The latter is the configuration eventually used in the setup, as alongside a more stable setup, the ideal fidelity laid between 6 and 7 brewster windows, which would make adding more windows useless change that.
Furthermore, the brewster window mount has two sections in which the layout of one half is mirrored to that of the other half. The reason behind this is that at each transmission through the brewster window, the optical beam path gets offset, which significantly reduces the coupling into the cavity fiber. In order to combat this, the other half of the brewster window placement was mirrored so that one half shifts the beam path up (relative to the direction of the into which the brewster window is pointing to come up with a better way to explain this), and the other shifts it back down. Although this is only the case for when the total amount of brewster windows is even and both halves are balanced, at most the shift would be equal to that of a single brewster window shifting the optical beam path, which is able to be compensating by realigning the input bea into the cavity fiber.
For the characterization of the brewster windows themselves, two measurements were conducted. For the first one, only a singular brewster plate was inserted into the custom mount to check the average attenuation across a set of brewster windows in the V polarization. This ensures that all brewster windows operate up to spec, and that the attenuation of the V polarization is happening in constant jumps. The second one, places multiple brewster windows in an array to closely resembe the setup used in the actual experiment. With this one can then used the measured powers in transmission both for H and V to determine the ideal amounts of brewster windows one would need to use in order to achieve the highest gate fidelity.

4.3 Photon Qubit Input-Output Path

For the atom-photon gate, weak coherent pulses are used in order to simulate the effect of single photons getting reflected from the cavity. It's therefore imperative to precisely control the power going into the cavity, and also the frequency which the light possesses. For that specific task the following setup was constructed:

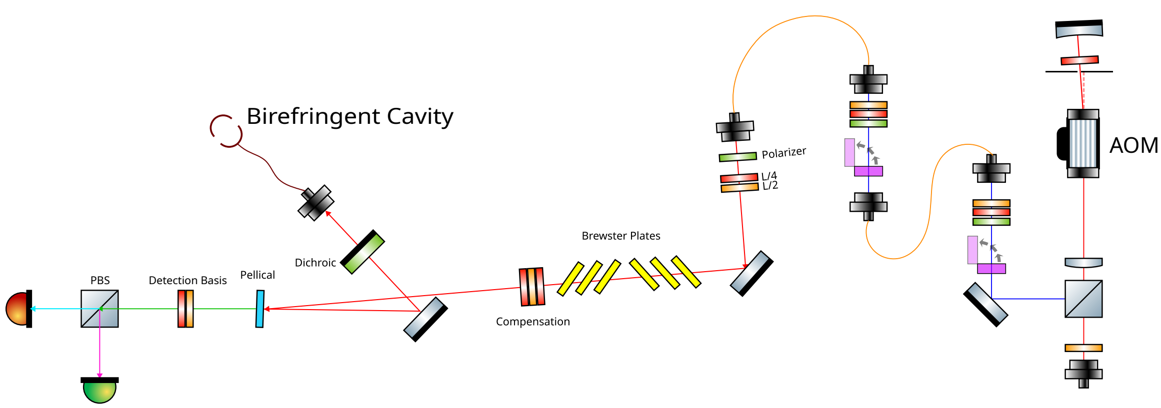

Figure ?? showcases a double-pass AOM track which allows to offset the frequency of the incoming light by two times the freuqency at which the crystal inside of the AOM is oscillating. The light going into the AOM is blue detuned to the unshifted D_2-line by include actual offset. Part of the laserlight (not shown here), can then be superimposed either with a stable reference frequency source like an optical freuqency comb from the company *MenloSystems GmbH* when the laser frequency doesn't need to be tuned, or with another laser, for when the frequency of the laser needs to be scanned, e.g. for performing a normal-mode spectroscopy to determine the coupling strength g. Regardless of what the other source may be, one can then measure the optical beat signal via conventional photo detectors, and use the beat signal frequency as a feedback for the current of the laser one wants to stabilize the frequency of. The beampath also features two stages connected via polarization maintaining fibers (PM), at which ND filters can be flipped into the path in order to reduce the power giving rise to the weak coherent pulses needed for the gate.

After the AOM track, the light is then coupled into a PM fiber, where it then leads to the so-called *atom table*, at which the atoms are trapped, positioned, and prepared via the aforementioned methods. At the beginning of the track two automatically controlled waveplates allow for the setting of an arbitrary polarization which then gets sent through a set of brewster windows. The reasoning behind he usage of the brewster plates will be explained later. Afterwards a set of compensation waveplates ensures that the H and V polarized light going into the system, are mapped to the π and V mode of the eigenmodes of the cavity. The light then gets reflected off a pellical, which transmits most of the incoming light and reflects a small portion of our light ($\approx 1\%$), onto a mirror which then gets reflected off of a mirror, into the fiber cavity coupler.

The reflected light coming from the cavity then gets reflected off of the previously mentioned mirror, transmitted through the pellical at minimal losses, and coupled into the detection setup. Here, one can set an arbitrary polarization measurement basis via another set of automatically controlled waveplates. It is thus possible to map the reflected light in two modes which then get detected via two SNSPDs.
Before the construction of the current beam path setup, the only way to couple light, other than that of the intracavity laser, into the cavity, was done via the so called *probe beam*. The light coming from the probe beam would be coupled into the cavity-fiber, through the transmission of one and subsequent reflection off of another dichroic mirror. Each of these mirrors is a include product names here from the company include company name here, and possess the feature of transmitting most of the incoming light and reflecting only a small portion. This setup was chosen so that both the intracavity light trap and the probe beam could be coupled into the fiber simaltaneously.

The problem with this way of coupling light into the cavity, for a beam where precise control over the polarization and it's losses is crucial, was the fact that in reflection the dichroic mirrors were introducing additional polarization dependent losses in the s and p polarization. An earlier setup chose to still couple light into the cavity when reflecting off of the dichroic mirror, while including a set of compensation waveplates to map the H and V polarization of the incoming light to the s and p polarizations of the dichroic mirror, and then another set of compensation waveplates right in front of the fiber cavity coupler to map s and p to π and V. However with no means to check the polarization of the reflected light from the dichroic mirror via a polarimeter, due to special constraints this proved to be unreliable.

Finally by changing the placement of the beam coupler for the beam, and instead opting to use a pellicle, which had much smaller polarization dependent losses in reflection, one was able to circumvent the issue. Furthermore, one was able to very that the polarization dependent changes both in reflection and transmission through the pellical were minimal, thus ensuring that the correctly reflected light won't receive further penalties when going to the detectors.
include numbers that showcase the losses for each polarization and stuff for the dichroic mirrors and also for the pellical (which are much smaller but still).

Chapter 5

Atom-Photon Gate

5.1 Experimental Parameters relevant for the gate mechanism

5.1.1 Cavity coupling and losses

5.1.2 Brewster plate attenuation

5.2 Atom-Photon gate performance

5.2.1 Gate Truth Tables

5.2.2 Generation of Bell States

Chapter 6

Summary and Outlook

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